Digital Video General Concepts

Video generalities

- Video is a sequence of frames consecutively transmitted and displayed so to provide a continuum of actions.
- This is obtained by adjusting the frequency of frames to the properties of the visual human system

- We are accustomed to watching videos that are played at a 24-30 fps rate. At these rates human eyes and brain process the visual input as one continual stream of information
- The maximum number of frames per second that a person could see is anyway credited to be higher: some research suggests that human brain might actually be able process an image that eyes see for only 13 milliseconds i.e. process about 75 frames per second….

Analog and digital video

- *Analog video* is a video signal transferred by analog signal. It contains the luminance (brightness) and chrominance (color) of the image. No more in use in Italy from 2012.

- *Digital video* comprises a series of digital images displayed in rapid succession. Digital video was introduced commercially in 1986 with the Sony D1 format and Sony's Digital Betacam and was hence widespreaded with computers. In addition to uncompressed formats, popular compressed digital video formats today include H.264 and H.265.

- There are advantages with digital video:
  - Digital video can be copied with no degradation in quality. No matter how many generations of a digital source is copied
  - Digital video can be manipulated and edited on a computer-based device. More and more, consumer-grade computer hardware and software are available.
  - Recording digital video is very inexpensive. Digital video increased in quality with the introduction of advanced compression standards

Digital video in use

- Digital television including higher quality HDTV started to spread in most developed countries in early 2000s.
- Digital video is increasingly diffused in film industry. Paramount has been the first to produce only digital, from 2013
- Digital video is used for Internet distribution of media, including streaming video and peer-to-peer movie distribution. Digital video is also used in modern mobile phones.

Digital video

- Distinguishing aspects of digital video are:
  - Capture formats
  - Color spaces
  - Encoding
  - Resolution
  - Aspect ratio
  - Bitrate
  - Compression
  - Sampling
  - Compression standards

Digital video capture formats

- Most video cameras available on the consumer market are digital video cameras. A digital video camera is a device that captures motion picture information from live environments, encoding it into data that can be decoded into electronic visual media.

- Digital video cameras may come in two different image capture formats: *interlaced scan* and *progressive scan* :
  - *Interlaced cameras* record the image in alternating sets of lines: the odd-numbered lines are scanned first, followed by the even-numbered lines. One set of odd or even lines is referred to as a *field*, and a consecutive pairing of two fields of opposite parity is called a *frame*.
  - *Progressive scan cameras* record each frame as distinct, with all scan lines being captured at the same moment in time.

Fields

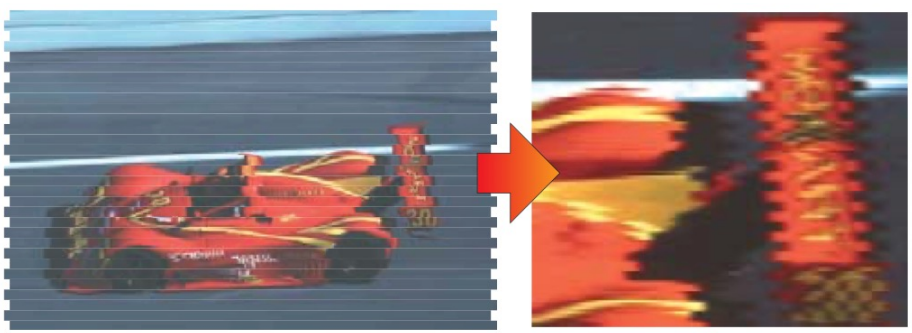
Interlaced scan

Progressive scan

Interlaced video

- Interlacing was an early form of video compression that was used to make the video look smoother to the human eye while being able to send less data related to the broadcast. If an interlaced video has a frame rate of 30 frames per second the field rate is 60 fields per second.

- With digital video compression, interlacing introduces inefficiencies:
  - Because each interlaced video frame is two fields captured at different times, frames can exhibit motion artifacts known as *interlacing effects*, or *combing*, if recorded objects move fast enough to be in different positions when each individual field is captured
  - when the subject contains vertical detail that approaches the horizontal resolution of the video format (a finely striped jacket on a news anchor) there can be *interline twitter* like a shimmering effect



Interlaced video today

- Cameras:
  - some modern digital cameras still offer interlaced mode.

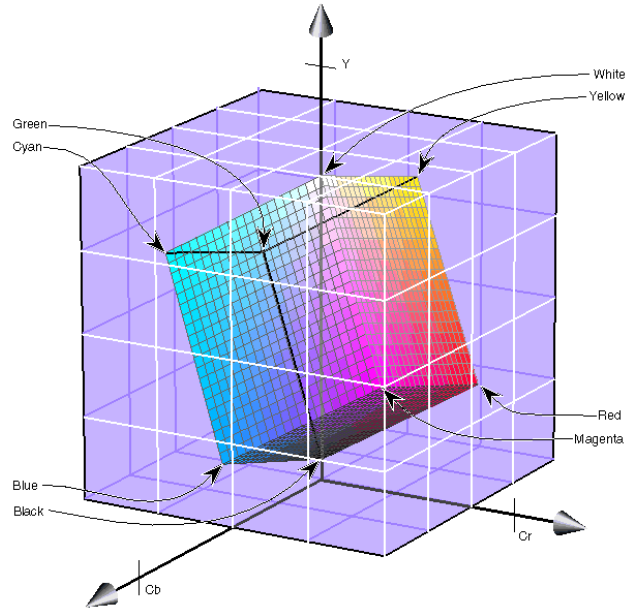
- Monitors:
  - LCD monitors and flat panels TV support progressive display. To display interlaced video on a progressive display requires a process called deinterlacing. EBU European Broadcasting Union promoted HDTV 1080p 50 progressive to replace 1080i 50
  - Most modern computer monitors do not support interlaced video. Playing back interlaced video on a computer display requires some form of deinterlacing in the software player

Digital video color spaces

- Video color is displayed in RGB (monitors use RGB). Although RGB color components could be used to represent color information in video however these signals are expensive to record, process and transmit.
- Digital video is therefore transmitted and stored using YCbCr. or Y'CbCr. *color spaces* that distinguish instead *brightness* and *chrominance* information (without / with gamma encoding).

- With YCbCr. or Y'CbCr. the values are scaled and rounded, and offsets are added

$$
\begin{aligned}
Y' &= 16 + \frac{65.738 \cdot R'_D}{256} + \frac{129.057 \cdot G'_D}{256} + \frac{25.064 \cdot B'_D}{256} \\
C_B &= 128 - \frac{37.945 \cdot R'_D}{256} - \frac{74.494 \cdot G'_D}{256} + \frac{112.439 \cdot B'_D}{256} \\
C_R &= 128 + \frac{112.439 \cdot R'_D}{256} - \frac{94.154 \cdot G'_D}{256} - \frac{18.285 \cdot B'_D}{256}
\end{aligned}
$$



Digital video encoding

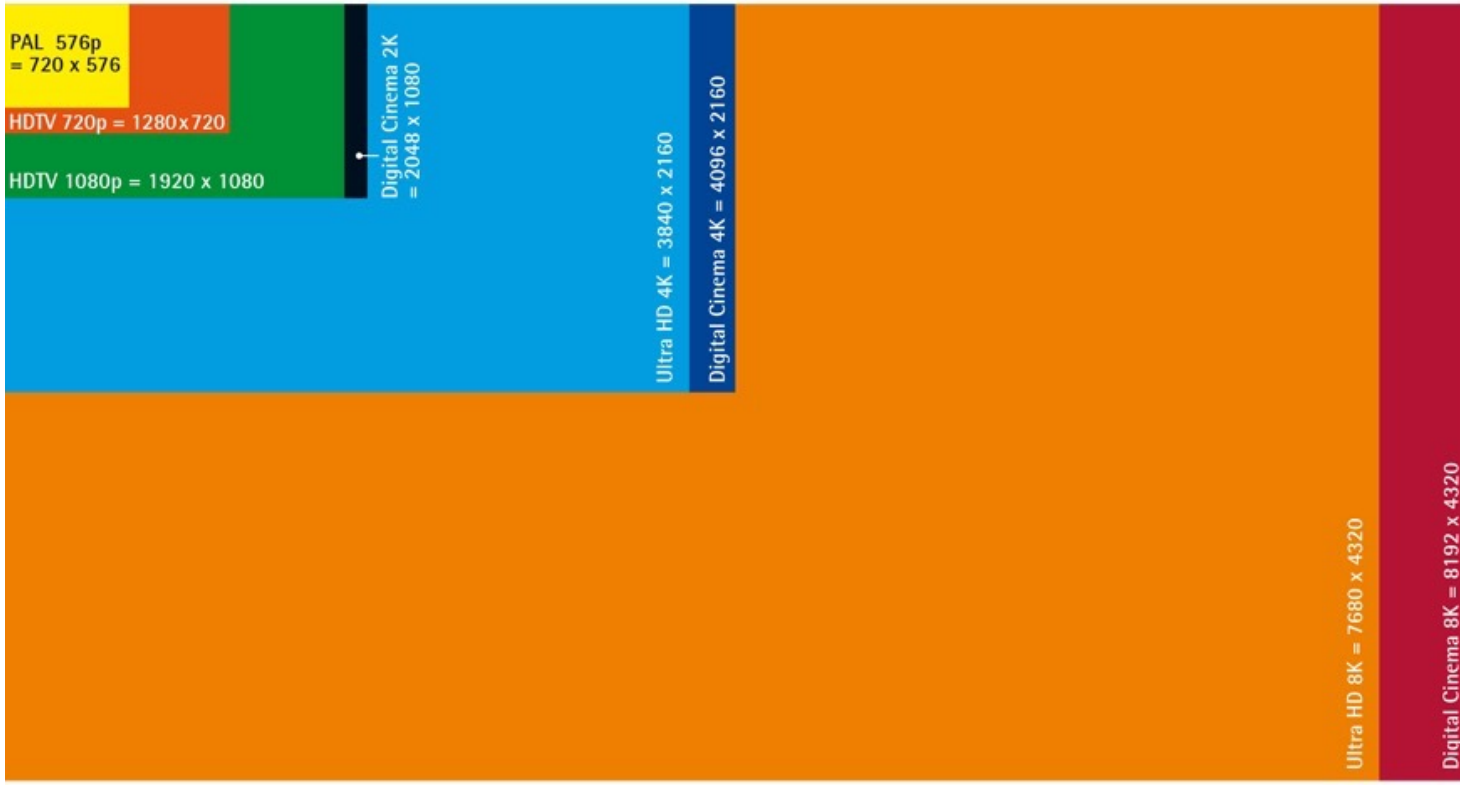
- Brightness and chrominance of images can be carried either combined in one channel as in *composite encoding* (brightness and chrominance information are mixed together in a single signal) or in separate channels as *component encoding*.
  - Analog video used either *composite* or *component encoding*
  - Digital video uses *component color encoding*

Digital video resolution

- Nowadays, digital television broadcasts via satellite, terrestrial, online and any other video content are available in different image resolutions:

  - **SD (Standard Definition) 576p**: *720 × 576 pixel* (if in 4:3), *1.024×576 pixel* (if in 16:9 native).
    Still used by DTT/SAT TV channels. A good option on very slow internet connection speeds.
  - **HD (High Definition) 720p**: *1280 × 720 pixel*.
    Generally used for satellite broadcasts and terrestrial TV (e.g. Sky TV, Sky Sport …)
  - **HD 1080i (Full HD)** : *1.920 × 1.080 pixel.*
    Provides 1.080 lines in interlaced mode. Preferred format by satellite broadcasts and terrestrial TV.
  - **HD 1080p (Full HD)** : *1.920 × 1.080 pixel.*
    Provides 1.080 lines in progressive mode. Used for video storage on physical support and online streaming on demand (YouTube, Netflix, Amazon Prime Video, ….).
  - **ULTRA HD 4K**: *3840 × 2160 pixels for television, 4,096 × 2,160 pixels for cinema.*
    State of the art of television broadcasting and very high definition video content.
  - **ULTRA HD 8K**: *7.680 × 4.320 pixels for television, 8.192 x 4.320 for cinema*.
    Next future very high quality. Currently available only for test broadcasts via satellite and on 5G mobile networks for sport events

- 4K (Ultra HD) is going to be the standard for video production. Google, YouTube, Vimeo support upload of 4K video
- 4K video can be useful when we need a crop in high resolution of a large image. But it requires wideband transmission



Digital video aspect ratio

- The *aspect ratio* of an image describes the proportional relationship between its width and its height.

- Current standards for digital video *aspect ratio* are:
  - **4:3** (1.33:1) for standard television has been in use since the invention of moving picture cameras and many computer monitors used to employ the same aspect ratio.
  - **16:9** the international standard format of HDTV. Many digital video cameras have the capability to record in 16:9
- **Square video and vertical video** are new video formats more suited to mobile devices.

- Square video (up to (1200 x 1200p) was popularized by mobile apps Instagram and then supported by Facebook and Twitter.
- Vertical video (1080 x 1920p with 9:16 format) was popularized by Snapchat and is also adopted by Twitter and Facebook.

Digital video bitrate

- *Bandwidth* is the frequency range of the video signal measured in MHz. The higher the bandwidth is the more information is carried on.

- For digital video we use the term *bitrate,* counting the number of bits that are conveyed or processed per unit of time (measured in bits per second) as the equivalent of bandwidth:
  - 16 Kbit/s videophone quality (talking heads)
  - 128-364 Kbit/s videoconferencing quality with video compression
  - 1.25 Mbit/s video CD quality with MPEG1 compression
  - 5 Mbit/s DVD quality with MPEG2 compression
  - **8-16 Mbit/s HDTV quality with MPEG4 compression**
  - **29.4 Mbit/s FULL HD quality**
  - 50 Mbit/s 4K ULTRA HD quality
  - …….

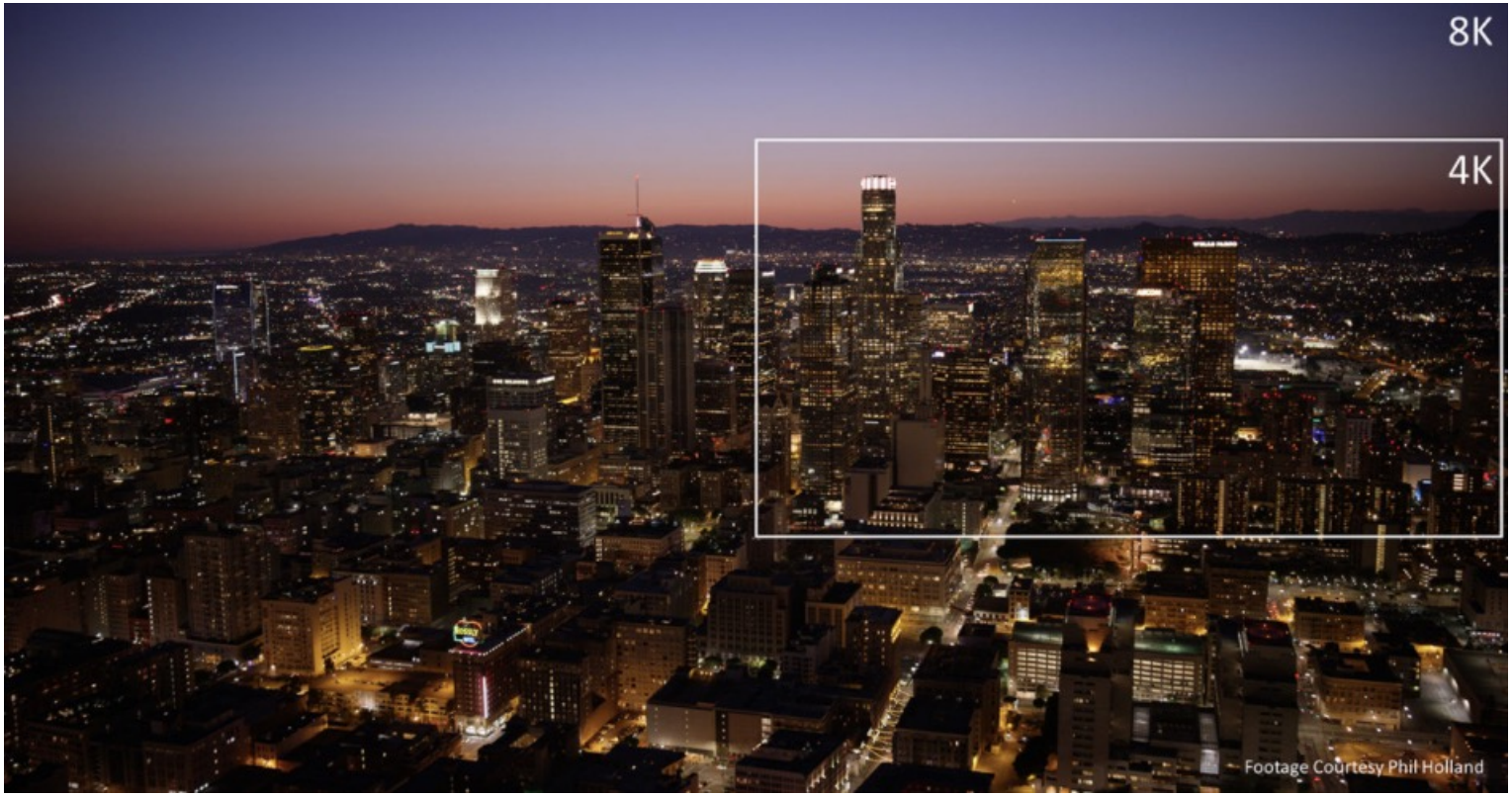
Constant or variable bitrate

- Video can be transmitted at *constant* or *variable* bitrate
  - *Constant bitrate* (CBR) maintains the same bit rate of transmission for the entire video. CBR can produce uneven quality. The variation in data density is handled by compressing the more intense parts of the video more than the simpler parts.
  - *Variable bitrate* (VBR) is a strategy to maximize the visual video quality and minimize the bitrate. On fast-motion scenes, a variable bitrate uses more bits than it does on slow-motion scenes of similar duration, yet achieves a consistent visual quality . VBR is the better choice for live streaming and for streaming video in general.

Bitrate and resolution

- There's a reciprocal relationship between *bitrate* and *resolution*: resolution is the density of the video itself, while bitrate has to do with its transmission.

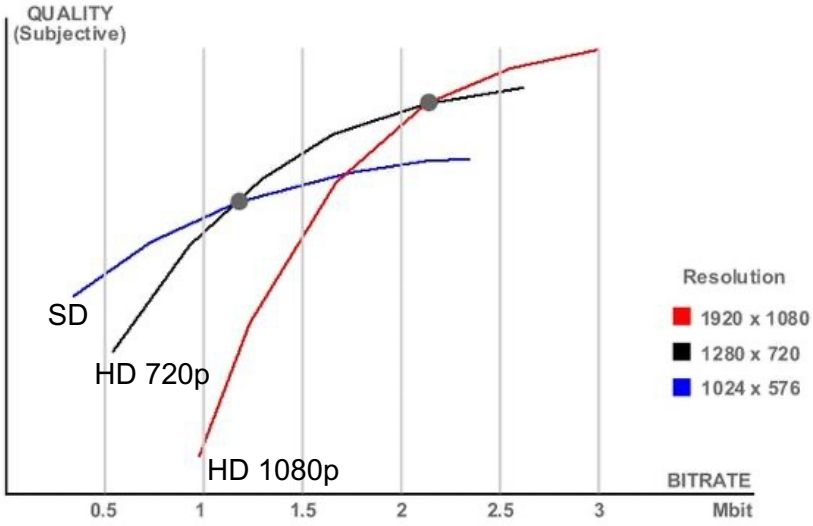
- A low bitrate can actually result in a poorer quality of high resolution video
- A higher bitrate allows transmission of higher-resolution video without as much loss of data.



Subjective viewing quality

- For effective video streaming, the video resolutions have to match with the right video bitrate.

Quality	Quality Resolution	Video Bitrate
LOW	480 X 270	400kbps
Medium	640 X 360	800 - 1200kbps
High	960 X 540/854 X 480	1200 - 1500kbps
HD	1280 x 720	1500 - 4000kbps
HD1080	1920 x 1080	4000 - 8000kbps
4K	3840 X 2160	8000- 14000kbps



Digital video compression

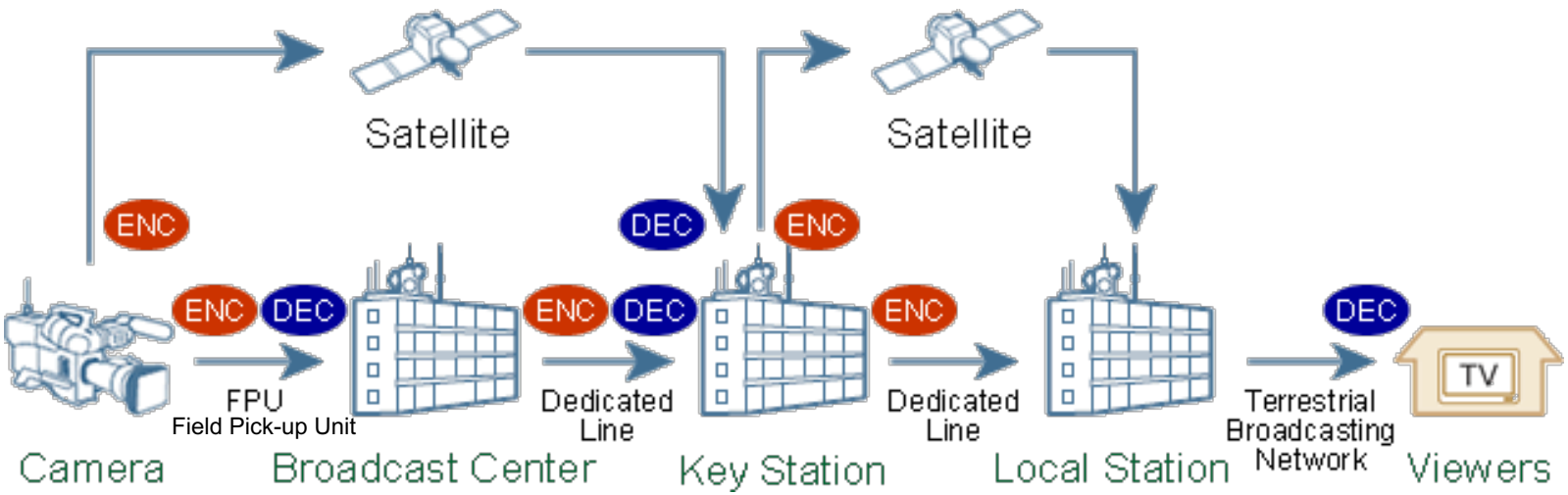
- Transmitting video always requires a certain amount of file compression. The higher the video resolution and the lower the bitrate, the more compression is required and the more data will be lost

- *Video compression* algorithms attempt to reduce the amount of information that is contained in video while preserving quality. Typically are lossy, starting with color subsampling

- A *codec* – or *coder/decoder* – is an encoding tool that processes video and stores it in a stream of bytes. Codecs use algorithms to effectively shrink the size of the video file, and then decompress it when needed.
  - Algorithms can be symmetric or not symmetric, in terms of (de)compression time/complexity. Typically, video compression algorithms for video distribution are highly asymmetric
  - Compression is both spatial and temporal i.e. both remove spatially redundant data (as in JPEG) and temporally redundant data

Compression cascading

- For the live broadcasting of sports or other events from the venue itself video codec systems are generally connected in cascade for multiple encoding and decoding
- To limit degradation in the quality of the video images delivered to viewers, it is essential that any image quality degradation that may be caused by the use of cascaded codec systems be minimized.



Courtesy of NTT Electronics

Digital video sampling

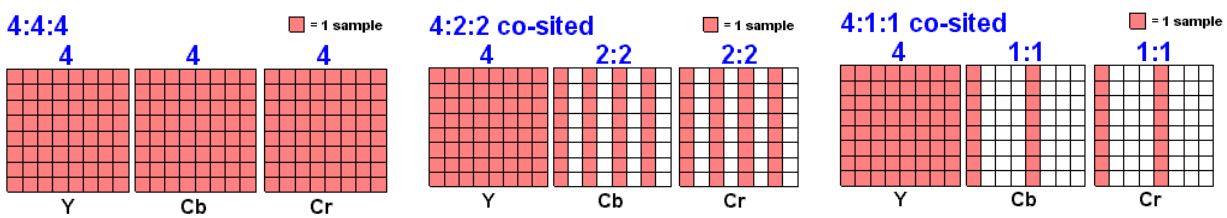
- Sampling is the simplest mechanism for data compression in video. It applies to luminance and chroma information in each video frame. Because the human visual system is less sensitive to the position and motion of color than luminance, bandwidth can be optimized by storing more luminance detail than color detail.

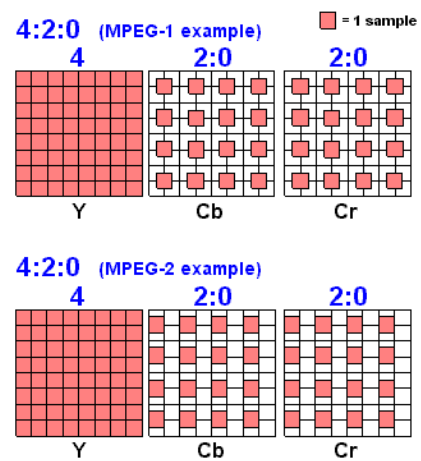
- Sampling is expressed with three values: x,y,z
  - x = relative number of luminance (Y) samples (sampling reference usually 4)
  - y = number of chroma (CbCr) samples for odd lines (in the first row of x pixels)
  - z = number of chroma (CbCr) samples for even lines (in the second row of x pixels)

  Es. 4:2:2 means that every 4 samples of luminance, there are 2 chroma samples both in the odd and the even lines. It compresses frames as it drops data.
  4:2:0 means that every 4 samples of luminance, there are 2 chroma samples in the odd lines and and 0 chroma samples in the even lines. It provides higher compression

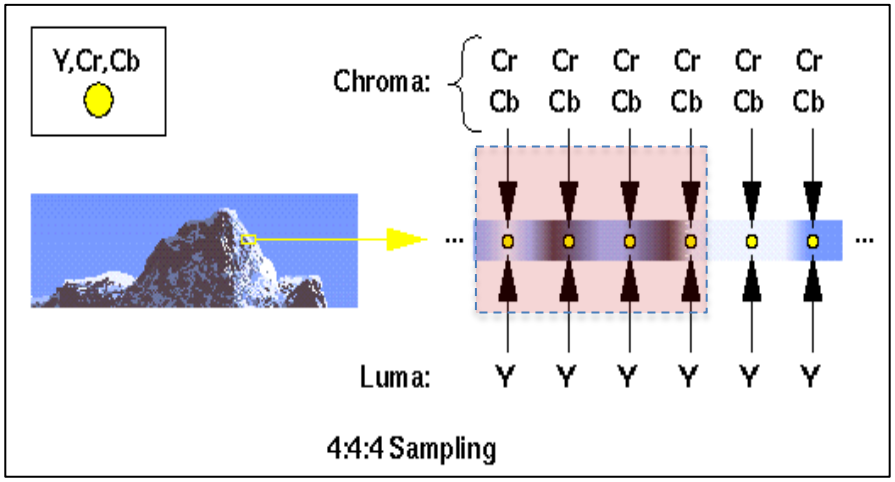
Sampling rates

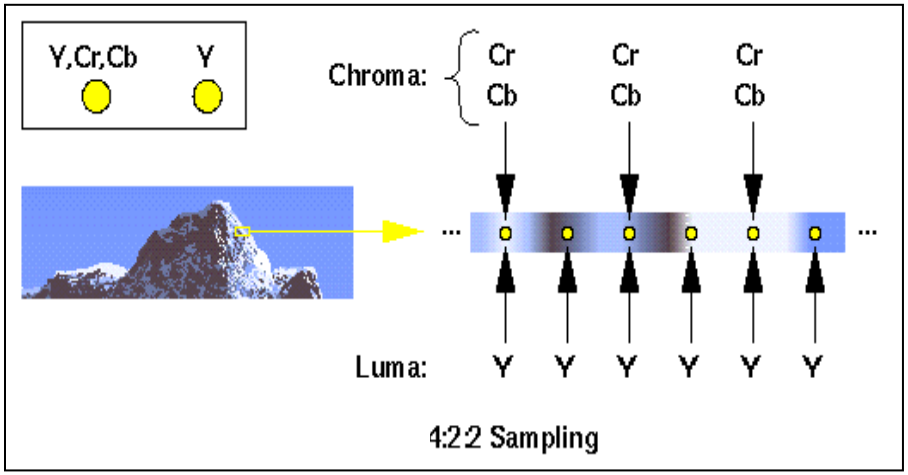
- Digital video can be stored and transmitted with different sampling rates:
  - 4:4:4 (Cb/Cr same as Luminance) Cb and Cr are sampled at the same full rate as the Y. MPEG-2 and H.264 support 4:4:4 coding. When video is converted from one color space to another, it is often resampled to 4:4:4 first.
  - 4:2:2 (color samples are 1/2 Luminance samples) Cb and Cr are sampled at half the horizontal resolution of Y and taken at the same time as Y. It is considered very high quality and used for professional digital video recording. It is an option in MPEG-2 and H.264.
  - 4:1:1 (color samples are 1/4 Luminance samples) Cb and Cr are sampled at one quarter the horizontal resolution of Y and taken at the same time as Y.
  - 4:2:0 (color samples are 1/4 Luminance samples) Cb and Cr are sampled at half the horizontal and vertical resolution of Y. MPEG-1 and MPEG-2 use 4:2:0, but the samples are taken at different intervals. AVC/H.264 also uses 4:2:0.

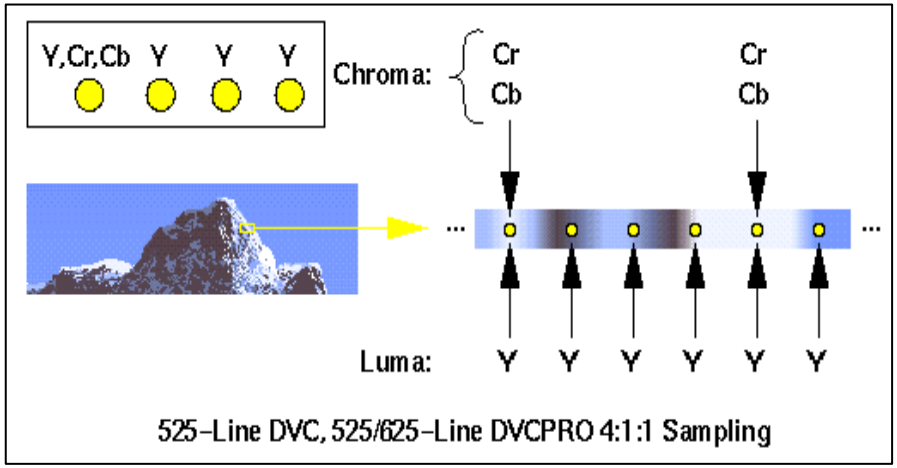


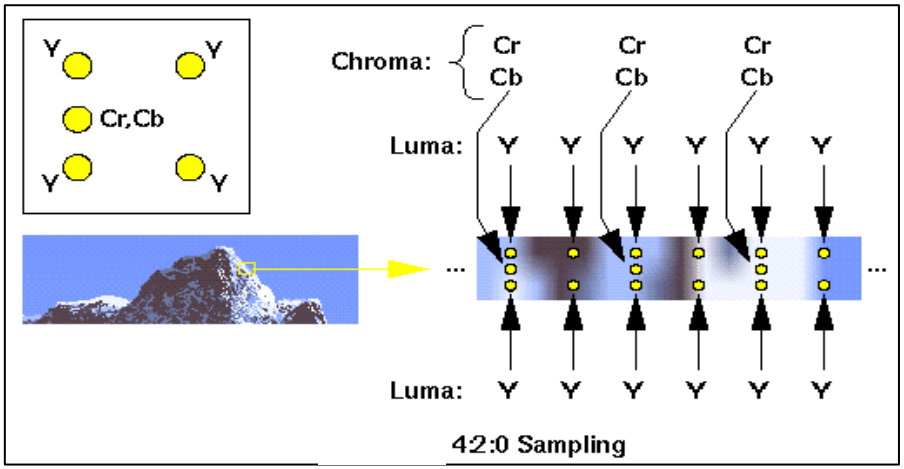


Digital video sampling



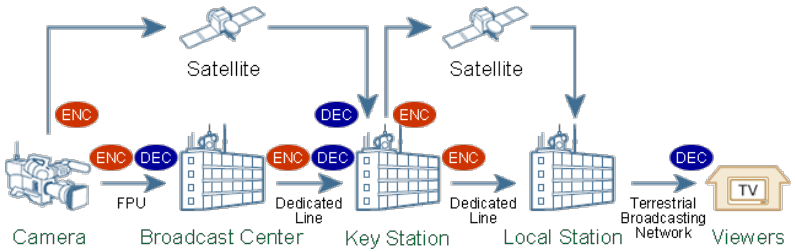






Effects of sampling

Difference in image quality between 4:2:2 and 4:2:0 chroma sampling of the AVC/H.264



Original

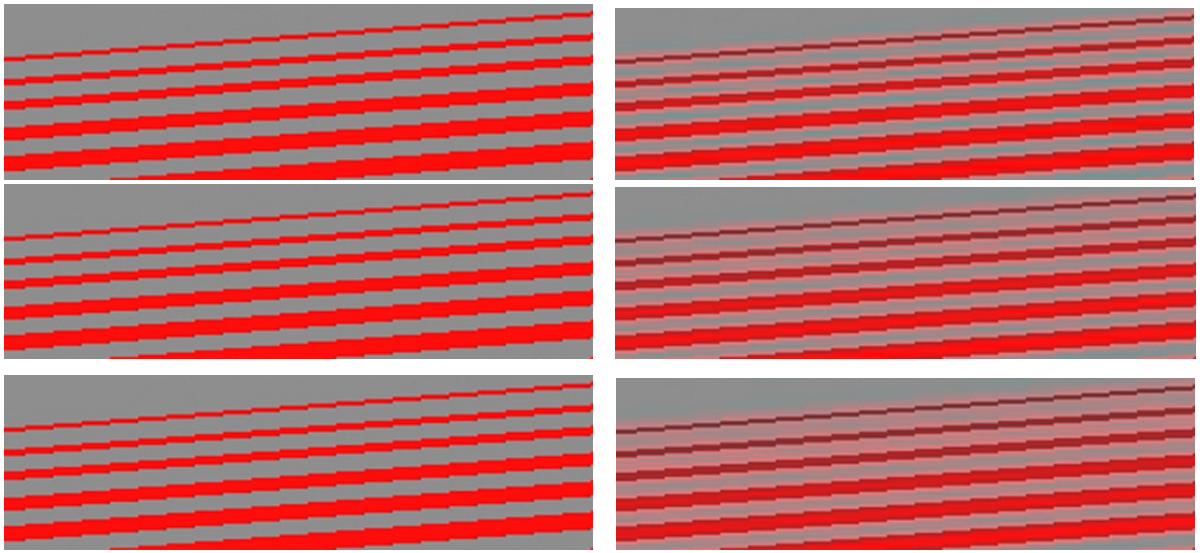
4:2:2 sampling

4:2:0 sampling

1st codec

2nd codec

3rd codec



Courtesy of NTT Electronics

Compression goals

- Suppose we have a video with a duration of 1 hour (3600sec), a frame size of 640x480 pixels at a color depth of 24bits (8bits x 3 channels) and a frame rate of 25fps.

- This example video has the following properties:
  - pixels per frame = 640 * 480 = 307,200
  - bits per frame = 307,200 * 24 = 7,372,800 = 7.37Mbits
  - bit rate = 7.37 * 25 = 184.25Mbits/sec
  - video size = 184Mbits/sec * 3600sec = 662,400Mbits = 82,800Mbytes = 82.8Gbytes

- When compressing video we aim at reducing the average bits per pixel (bpp):
  - with chroma subsampling we reduce from 24 to 16-12 bpp
  - with JPEG compression we reduce to 8-4 bpp
  - with MPEG codecs we go below 1 bpp

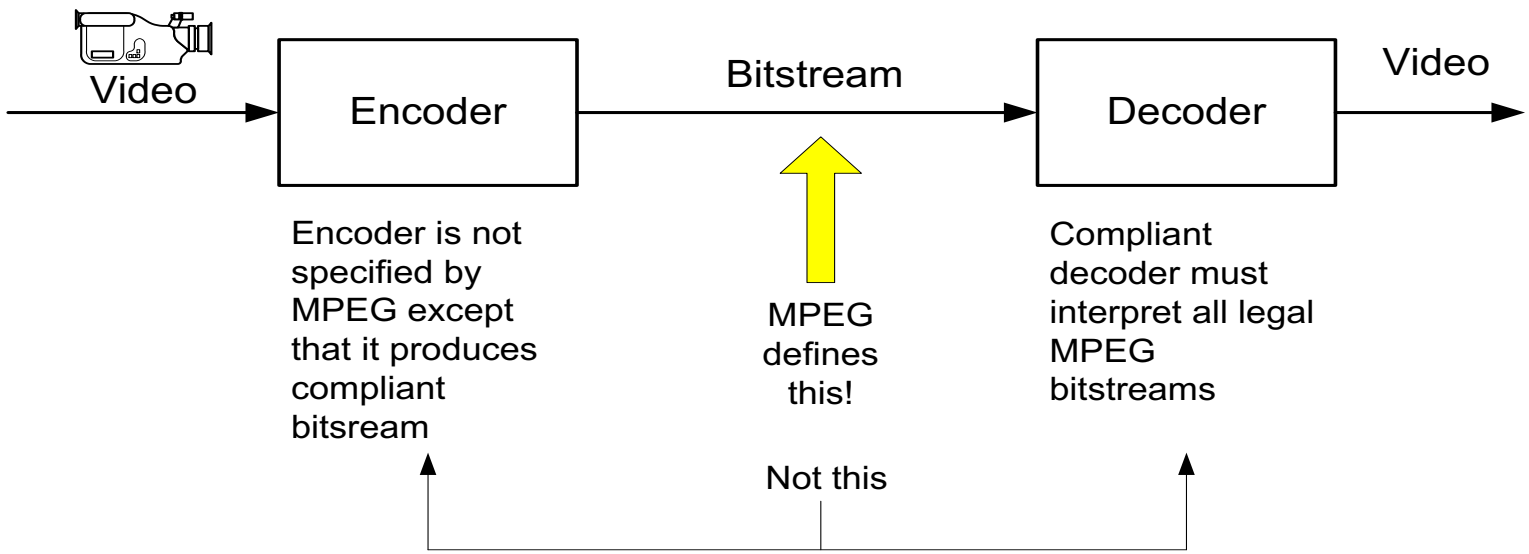
Compression standards

- Compression standards have been originated by two experts groups: the MPEG (Moving Pictures Experts Group) and the ITU-T VCEG (Video Coding Expert Group). MPEG specialized in broadcast (television), while the VCEG focused on telecommunications (phone, internet). Today their goals mostly overlap as everything is going the way of the Internet .

- In 1988, H.261 was created by the VCEG, mainly for ISDN/Videoconferencing work. It had a maximum bit rate of 2 Mbps, and was limited with a chroma sub-sampling of 4:2:0. In the same time JPEG became a popular codec for images.
- In 1993 the MPEG following the broadcast industry's needs adopted H.261 and JPEG and defined the MPEG-1 suite. It was limited to 1.5 Mbps, 4:2:0 and stereo audio only.
- Thereafter VCEG and MPEG developed together….

MPEG compression

- MPEG compression methods have been defined according to ISO standard and aim at reducing redundancy in video sequences:
  - The MPEG defines the protocol of the bitstream between the encoder and the decoder
  - The encoder is left to the designer. The decoder is defined by implication.



Progress of MPEG standard

The story of two groups - MPEG and VCEG						
Year	**MPEG**	**Part**	**Layer/Profile/Type**	**Usage**	**VCEG**	Variants
1984	Not formed	Practically not useful			H.120	
1988	Not formed	Videoconferencing			H.261	
1993	MPEG-1	VHS and Television Recording				
		Part 1	Systems			
		Part 2	Video	VCD	H.261	
		Part 3	Audio			
			Layer 1			
			Layer II			
			Layer III	**MP3**		
1999	**MPEG-2**	Broadcast, Distribution, DVD				
		Part 1	Systems			
			Program Stream			
			Transport Stream			
		Part 2	Video		H.262	HDV, XDCAM
		Part 3	Audio			
			Layer 1			
			Layer II			
			Layer III	**MP3**		
2004	**MPEG-4**	Broadcast, Internet, Blu-ray				
		Part 1	Systems			
		Part 2	Video		H.263	HDCAM SR
		Part 3	Audio			
		Part 10	Advanced Video Coding	**MPEG-4 AVC**	**H.264**	AVCHD, XAVC
		Part 14	MP4 Container	**MP4**		
2013	MPEG-H	Part 2	Video	**HEVC**	**H.265**	

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MPEG-1 H261, MPEG-2 H262, MPEG-4 H264

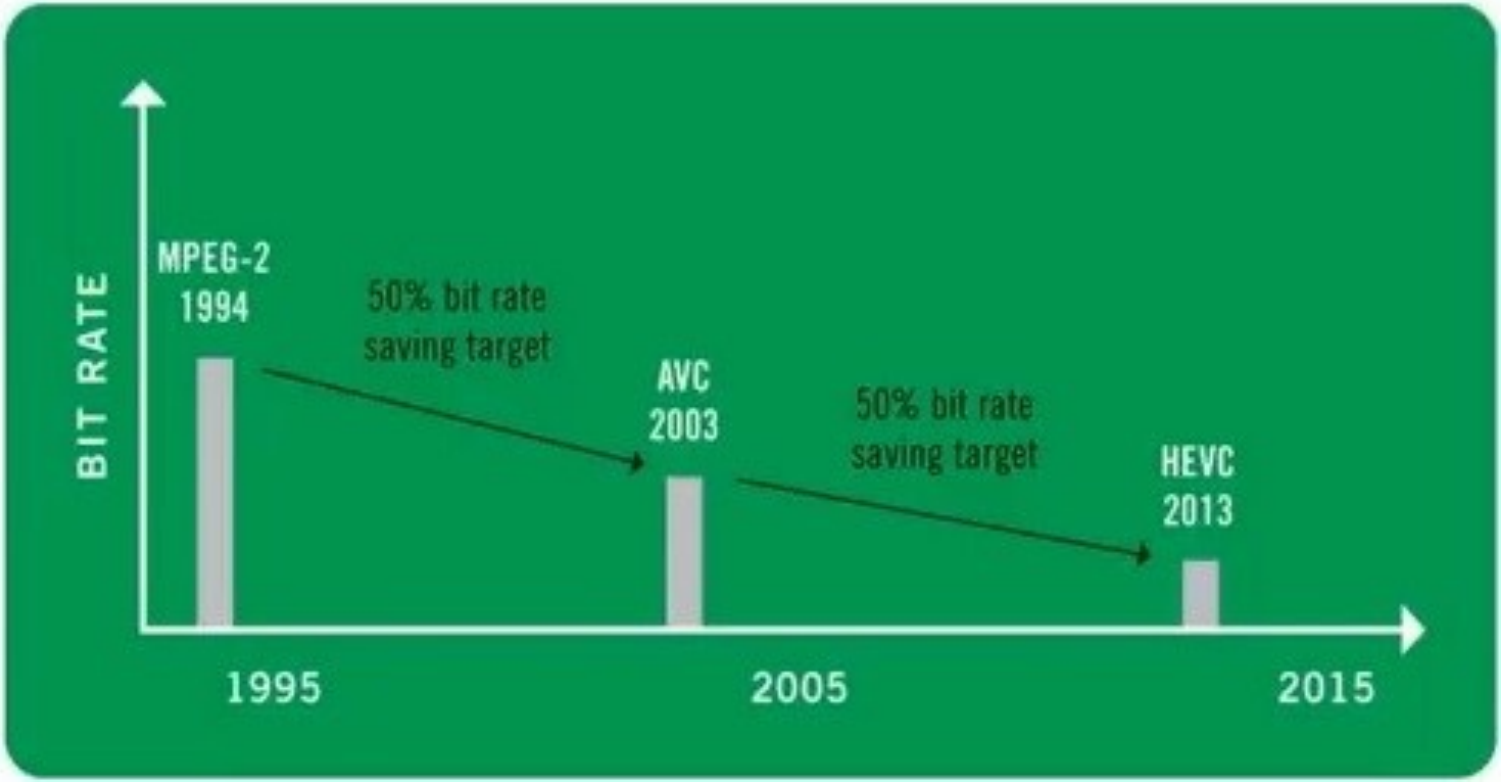
- MPEG-1 H261 No more in use
  - VHS Quality at 1.5 MBits/s
  - Basis of Video-CD
  - MP3 audio

- MPEG-2 H262 Still used for broadcast. Dismissed on Digital Terrestrial TV in Italy from March 8, 2022
  - designed for broadcasting and storage
  - Bitrates: 4-9 MBits/s
  - Satellite TV, DVD

- MPEG-4 part 10 H264 The most widely used codec today
  - encoding method for devices with limited resources (media players and mobile phones)
  - Internet and very low-bitrate transmission: 4Kbps to 240Mbps (Part 10/H.264)
  - Drives the Blu-ray disc
  - Coding of media objects

MPEG-H H265

- MPEG-H H.265 or HEVC (High Efficiency Video Coding). Key standard for videosurveillance with Full HD e 4K cameras. There are many vendors already compatibile with this format.
  - Support up to ULTRA HD 8K
  - File size half the size of H.264 with better quality



Video containers

- Video is available to users in *containers*. Containers are like an envelop that contains compressed video data. A container has the purpose of bundling all of the audio, video, and codec files into one organized package.
- The container often also contains chapter information for DVD or Blu-ray movies, metadata, subtitles, and/or additional audio files such as different spoken languages.

- Containers must be distinguished from codecs.

Containers

- Most popular containers are:

  – **MKV** (open source)

    MKV is a rapidly growing format. The container itself supports almost any audio or video format. However, they are not often supported from editing software and on some operating systems an extra playback program is required.

  – **MP4**

    MP4 is the recommended format for uploading video to the web, and services such as Vimeo and YouTube have it listed as their preferred format. The MP4 container utilizes MPEG-4 encoding, or H.264, as well as AAC or AC3 for audio. It is widely supported on most consumer devices, and is the most common container used for online video.

  – **AVI**, **WMV**, **ASF** (Microsoft)

    The **AVI** container is probably the best known one, but is also very old and cannot handle modern codecs like H. 264. **WMV** and **ASF** containers are not very flexible. If the video is played on a non-Windows system it usually needs an extra video player.

  – **MOV** (Apple)

    The **MOV** container supports about all codecs, but some operating systems require an extra video player

  - **XAVC, XAVC-S**

    Used in recent sony cameras, uses H.264 compression for 4K video

  – **FLV**, **SWF** (Flash Video) Flash is an aged container